

SCIENCE AND TECHNOLOGY IN ANCIENT INDIAN TEXTS

Concept of Agni in the Ṛgveda

In the Light of the Physical Sciences

Soma Basu & R.K. Saha

School of Vedic Studies, Rabindra Bharati University, Kolkata



Agni — God of Fire

Introduction

The aim of this paper is to examine scientific elements embedded in the mantras of the Ṛgveda without distorting their true purport.

As Acharya Brajendranath Seal warned: we must bring out scientific elements in whatever rudimentary form they appear — by keeping intact the true spirit of the text.



"A true historical perspective can only be acquired by historico-comparative studies with the application of the correct historico-comparative method." — Acharya Seal

India's Scientific Heritage in the Ancient World

Mathematics, Geometry, Trigonometry, Astronomy & Ecology

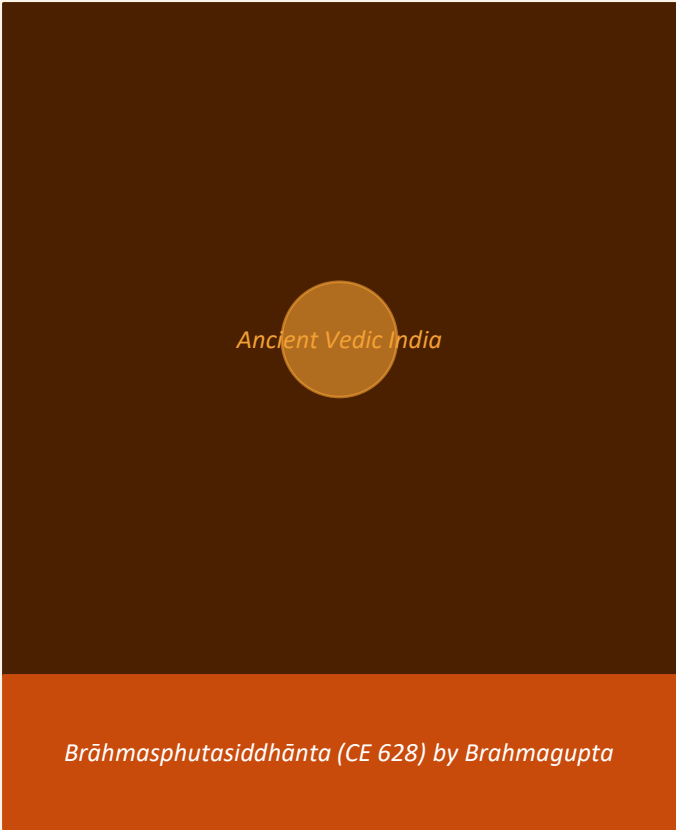
India contributed the earliest concepts in many branches of science, later discovered or brought to light in other parts of the world.

Architecture in the Vedic Age

ṚV 2.41.5 mentions a thousand-pillared palace. ṚV 10.99.3 references structures with hundred gates (śatadvārasya).

Medical Science

Caraka (1000 BCE) described antar indriya — the mind as the supreme organ that coordinates all senses and connects them to the brain.



Ancient Vedic India

Brāhmasphutasiddhānta (CE 628) by Brahmagupta

Vaiśeṣika School & Early Physics

Energy from Contrary Substances

The Vaiśeṣika School stated that the mixture of two ever-contradicted substances emits the most powerful energy.

The mixture of water and fire emits steam — a principle documented in India much earlier than James Watt's discovery of the steam engine.

This demonstrates that phenomenal growth of modern scientific culture began with deriving knowledge from the past — approached with a forward-looking scientific method.

Key Scientific Pioneers

Caraka

Medicine & Mind, ~1000 BCE

Brahmagupta

Mathematics, CE 598

Vaiśeṣika

Atomic Theory, Physics

James Watt

Steam Engine, CE 1769

The Ṛgveda: Science Embedded in Sacred Verse

Three Levels of Interpretation

ādhidaivika — Realm of deities; inner principles or power of nature

ādhibhautika — Realm of physical nature in all its aspects

ādhyātmika — The inner reality of man

Prof. Winternitz (History of Indian Literature, Vol. 1, p. 77):

"The songs of the Ṛgveda prove indisputably that the most prominent figures of mythology have preceded from personifications of the most striking natural phenomena."



Vedic fire sacrifice (Yajna)

01

Agni — Physical (Non-Divine) Attributes in the Ṛgveda

All That Is Warm Is Agni

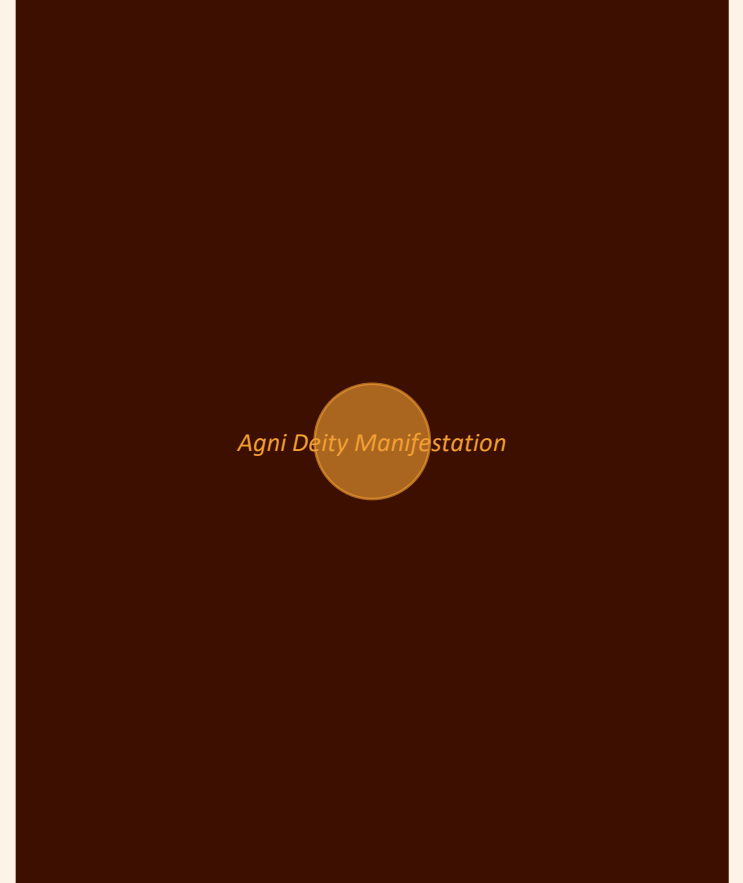
According to Vedic hymns, Agni manifests itself in natural phenomena as fire, heat and light. Wherever and in whatever form the principle of heat is present — that is Agni.

Key Sanskrit Epithets:

tapurjambhaḥ, tapuḥ, tapiṣṭhaḥ, tigmahete
tigmajambhas, tapiṣṭhena śociṣā, tigmaśocis

The Śatapatha Brāhmaṇa states:

"yac chuṣkam tad āgneyam" — What is dry relates to Agni



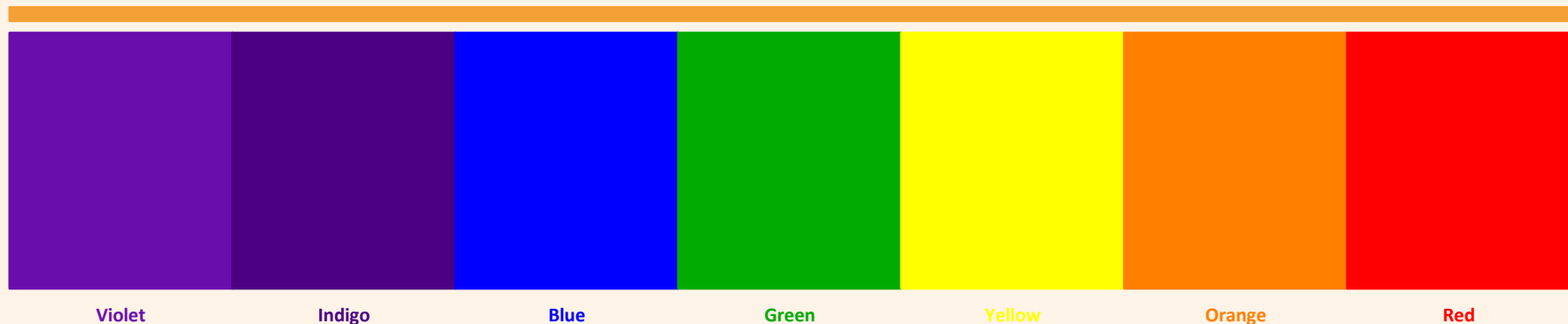
Agni as Seven-Rayed Light

In about 65 mantras visualised by 25 seers, Agni is praised as having different colours (rays) — with a maximum of seven.

ṚV 1.146.1: saptaraśmim — having/with seven rays

ṚV 2.5.2: Agni expressed as saptasamkhyākā raśmivad — 'in whom seven rays are manifested'

ṚV 4.7.5: sapta dhāmabhiḥ — Wilson translates as 'with seven flames' meaning seven different wavelengths



Visible light spectrum — seven colours as identified by Vedic seers millennia before Newton's prism experiments

Polychromatic Light: Vedic Seers & Newton

Various Epithets of Agni's Colours:

rohitā red-coloured

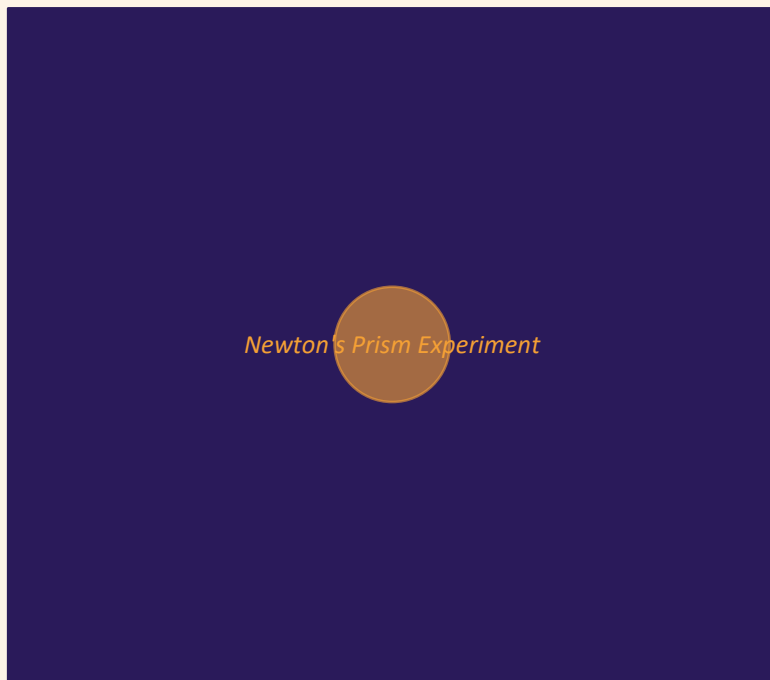
harikeśa tawny-rayed

nīlaprṣṭha purple dawn

hiranyarūpam golden-formed

sitiprṣṭha black-backed

These showed to Vedic seers that Agni (light/fire) was chromatic — composed of different colours and rays.



Newton's 17th century CE prism experiment proved unambiguously the meticulous observation and imagination of Vedic seers.

Agni as Supporter of the Universe

RV 5.7.6

Oft-quoted epithets — *dhartā*, *vidhartā*, *viśvasya dhāarakam* — meaning 'supporter of the universe'

Agni = Heat Energy everywhere

Modern physics (Rumford & Davy) proved that heat is a form of energy, proportional to absolute temperature. Every substance — solid, liquid, gas — consists of molecules always in motion.

Maxwell's Law (1859): In a dynamical system in thermal equilibrium, energy is equally distributed. Each degree of freedom has energy = $\frac{1}{2}kT$

Implication: For $T = 0$ (zero temperature), all motion stops. The universe cannot be sustained without heat — confirming the Vedic vision of Agni as the universal sustainer.

Kinetic Theory of Heat — Validating Vedic Insight

Heat = Kinetic Energy

Heat is a form of energy due to motion of molecules, proved by Rumford & Davy

Boltzmann Constant

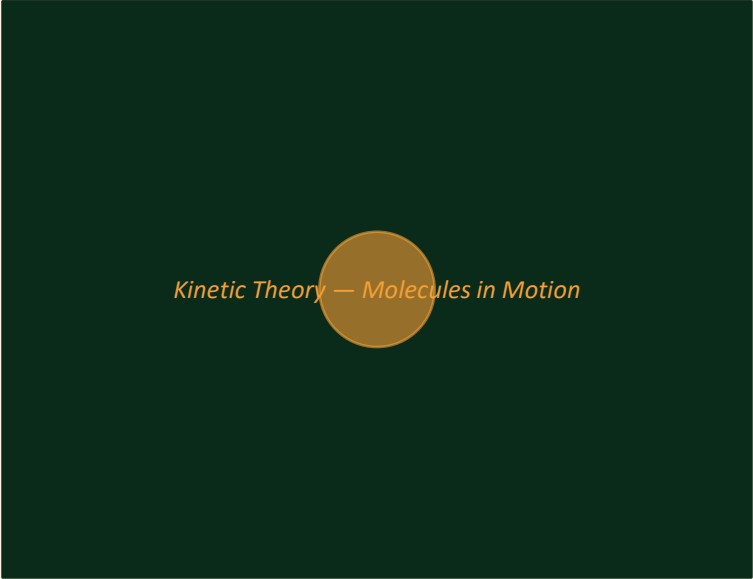
Energy per degree of freedom = $\frac{1}{2}kT$; 'k' is Boltzmann constant, T is absolute temperature

Equi-partition Law

Maxwell (1859): Energy equally distributed among all degrees of freedom in thermal equilibrium

Universal Presence

For the universe to exist, temperature T must always be positive — Agni must be present everywhere



Kinetic Theory — Molecules in Motion

Agni as Present Everywhere (RV 1.70.2)

*"garbho yo apām garbho vanānām garbhaśca sthātām garbhaś carathām | adrau cid asmā antardurone
viśām na viśvo amṛtaḥ svādhīḥ ||"*

— RV 1.70.2

Ten different seers in about 20 mantras visualised the presence of Agni everywhere:

- Within water • Within all moveable & immoveable things
- Cause of sustaining life • Responsible for growth of herbs, shrubs, plants
- Ripening of grains • Present in wombs of all living beings
- Responsible for digestion (jāṭharāgni)

02

Agni as Messenger & Light in Motion

Why Was Agni Addressed as 'Messenger' in 110 Mantras?

About 110 mantras by 34 seers praise Agni as messenger or oblation-bearer:

dyūtām (RV 1.44.11) — devānām dyūtasthānīyam

"Agni, the bearer of oblations, has become the messenger of the gods."

Why Agni over Indra or Maruts?

Modern science: Nothing travels faster than light.

- Speed of sound: 3.5×10^4 cm/sec
- Speed of light: 2.99×10^{10} cm/sec
- Light takes only ~8.25 seconds from Sun to Earth

Light travels without any medium, in all directions — celestial or terrestrial. Huygen (1690) proved light travels as waves in all directions through a universal medium (luminiferous ether).

Agni Travelling in All Directions & Most Rapid

Agni Travels Everywhere (RV 1.145.1)

17 seers in 36 mantras describe Agni as viśvāyuh — 'going everywhere'. Sāyaṇa: vyāptagamaṇaḥ, sa sarvatra gacchati.

RV 1.27.3, 1.140.5, 1.145.1

Agni is Most Rapid (RV 4.4.3)

Agni is ajiram, tūrṇitamo, urujrayasam — meaning 'most rapid'. Sāyaṇa: kṣipraṁ gantāraṁ, tvaritatamah, prabhūtavegam.

RV 3.9.8, 4.4.3, 5.8.6

Agni is All-Pervading (RV 6.15.8)

Agni is vigāhah, vibhuh — 'all-pervading' (sarvatra vyāptam). Light propagates simultaneously in all directions from a source.

RV 3.3.5, 6.15.8

03

Appendix: The Maruts & Principles of Statics

The Maruts — Storm Deities and Physical Forces

The Maruts (Storm Deity): next to Indra among atmospheric gods — celebrated independently in 33 hymns, conjointly with Indra in 7.

Physical Concepts in Vedic Descriptions:

Parallel Forces (Statics): abhi...mitho vapanta (ṚV 7.56.3) — 'they go together by their own pure paths'. Vedic seers always addressed Maruts in plural — mathematically correct because wind cannot act on a single point.

Resultant Force: yujā (ṚV 1.39.4) — 'May your collective strength be quickly exerted through the union of you' — the concept of resultant force.

Bending due to Couple: sthirāni namayiṣṇavaḥ (ṚV 8.20.1) — 'who can bend even what is firm' — Oldenberg: two unlike equal parallel forces causing bending.

Origin of Sound: Maruts set air-medium in vibrations while travelling — the same phenomenon behind sound generation in a flute.

Conclusion

Vedic seers, through meticulous observation and imagination, intuitively attained knowledge that took Europeans centuries to arrive at in modern science.

Key Scientific Insights Embedded in the Ṛgveda:

- Agni = fire/heat/light is universal — present everywhere (predates Kinetic Theory)
- Seven-rayed light — polychromatic nature (predates Newton's prism experiment)
- Agni as fastest entity, all-pervading — properties of light validated by modern physics
- Agni sustains the universe — consistent with Boltzmann's energy law
- Maruts encode concepts of parallel forces, resultant force, and origin of sound

"Thoughts of Vedic seers were scientifically significant which were lost in the midst of antiquity... the principles of fundamental science remaining embedded in whatever rudimentary form."